

# **CAB111:DATA VISUALIZATION USING PYTHON AND R**

L:0 T:0 P:4 Credits:2

**Course Outcomes:** Through this course students should be able to

CO1 :: recall the fundamental concepts and terminology of data visualization

CO2 :: explain the principles and best practices of data visualization

CO3 :: use Python and R libraries (such as Matplotlib, Seaborn, and ggplot2) to create basic plots and charts.

CO4 :: compare different data visualization techniques to determine the most appropriate method for a given dataset.

CO5 :: assess the quality and effectiveness of visualizations based on clarity, accuracy, and aesthetics.

CO6 :: develop complex, interactive visualizations that effectively communicate insights from data.

## **List of Practicals / Experiments:**

### **Visualization with Matplotlib library**

- Basic plots
- Line Plots
- Bar plot
- Histograms
- Scatter plot
- pie chart
- Area Plots
- Pie Charts
- Box Plots
- Bubble Plots
- Waffle Charts
- Word Clouds
- Pyplot in Matplotlib
- Line Plot
- Histogram
- Scatter
- 3D Plot
- Image
- Contour and Polar
- Matplotlib
- Axes Class
- axes () function
- add\_axes () function
- ax. Legend () function
- ax. plot () function

- sine and cosine functions.
- Multiple Subplot
- Create multiple subplots
- add title to subplots
- set single main title for all subplots
- turn off the axes for subplots Advance Data Visualization
- Visualizing the content of a 2D array
- Adding a colormap legend to figure
- Visualization nonuniform 2D data
- Visualizing contour lines
- Polar charts
- Plotting log charts for research

### **Creating Maps and Visualizing Geospatial Data**

- Introduction to Folium
- Maps with Markers
- Choropleth Maps Export Feature – Data Visualization
- Generating a PNG picture
- Generating PDF documents
- Multiple graphs plotting and export
- Inserting sub figure Hypothesis and Gradient Descent
- Understanding Hypothesis
- Implementation of hypothesis in python
- Gradient Descent
- Implement of gradient Descent

### **Visualization by using Seaborn Library**

- Relational plot
- Dist Plot
- Line Plot, Lmplot
- Categorical plot
- Stripplot
- Swarmplot
- Barplot
- Countplot
- Boxplot
- Violinplot
- Distribution plot
- Joinplot
- Distpot
- Pairplot
- Rugplot
- Regression plot

- Simple Linear plot with additional parameters (hue and markers)
- Setting size and color of the plot
- Displaying multiple plots
- Size and aspect ratio of plots
- Matrix plot
- Heatmaps
- Cluster Maps
- Style and Color
- Set the background to be white
- Set the background to be ticks
- Set the background to be darkgrid
- Set the background to be whitegrid
- Remove axes spine
- Despine Size and aspect
- Non grid plot
- Grid type plot Scale and Context
- Poster
- paper
- notebook and talk

#### **WORKING WITH R R basics**

- Math
- Variables and strings
- Vectors and Factors
- Vector operations
- Data Structure in R -Arrays & Matrices
- Lists & Data frames
- Conditions and Loops
- functions in R
- Objects & Classes
- Debugging
- R programming Fundamentals
- Conditions and loops,
- Objects and Classes
- Debugging Working with data in R - Reading CSV and Excel Files
- reading text files
- Writing and saving dataobjects to file in R
- String operations in R - Regular Expressions
- Dates in R
- Using Visualization tools – Bar Charts
- Histograms, Pie Charts
- Scatter Plots

- Line Plots
- Using Specialized visualization –Word Clouds
- Radar Charts
- Waffle Charts
- Box Plots
- Mapping - Creating Maps in R
- building interactive web pages
- Introduction to Shiny
- Creating and Customizing Shiny Apps
- Additional Shiny Features

#### **PYTHON AS TOOL**

- Crash course of Python
- Sample Scripts with Loops in python

#### **Data Visualization**

- Understanding Data Visualization, history and Architecture of Matplotlib

#### **Data Analysis**

- Understanding Data Analysis
- Exploring 1-D data
- Exploring 2-D data
- Bubble chart representation
- Data Munging

#### **Linear Algebra**

- What are vectors. various operations of vectors
- Understanding Matrices

#### **Statistics**

- Single set of data
- Concept of Central Tendencies
- Dispersion

#### **Probability**

- Probability concept
- Normal Distribution
- Central Limit Theorem

**Text Books:** 1. FUNDAMENTALS OF DATA VISUALIZATION by CLAUS O. WILKE, O Reilly Publishing

**References:** 1. DATA VISUALIZATION USING PYTHON PROGRAMMING by DR.ABHINAV, A P Publisher

